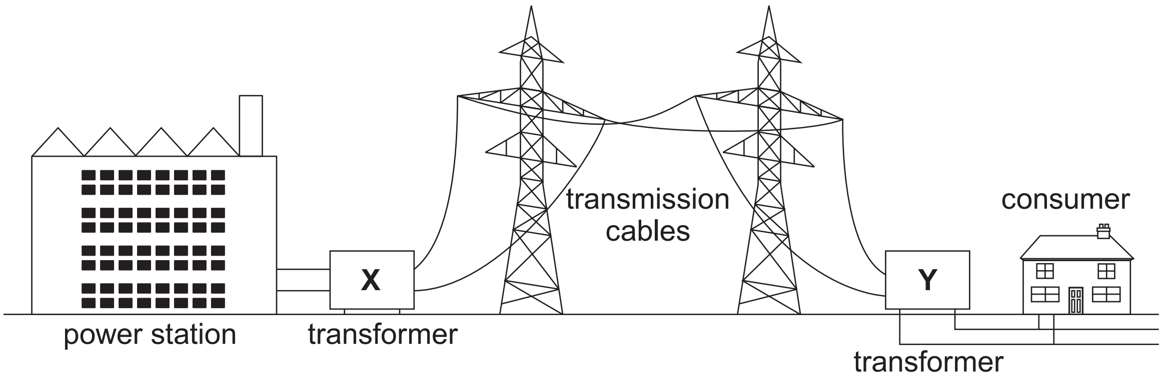
3. A diagram of part of the National Grid is shown below.



(a) Describe the role of the National Grid. [2]

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(b) Explain why transformers **X and Y** are used in the National Grid. [4]

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Examiner only

(c) Not all consumers can access the National Grid. An isolated farm with 5 sheds, in a rural part of mid-Wales is not connected to the National Grid.

The farmer needs to generate electricity for his sheds and he chooses a biomass generator that uses wood chips and pellets as an energy source and has a capacity of 100 kW.

Each of the sheds is supplied with 230 V with a maximum current of 80 A.

The farmer thinks that this generator will supply enough power for his 5 sheds.

(i) Use the equation:

$$\text{current} = \frac{\text{power}}{\text{voltage}}$$

to determine if the 100 kW generator will be able to supply the 5 sheds. [4]

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(ii) Explain the environmental advantages of using the biomass generator. [3]

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